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Environmental Chemistry (CY1010), FINAL EXAM (Marks 50)

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Answer all the questions

1. Which pollutant is mainly formed during high-temperature combustion and contributes to smog, respiratory problems, and ecosystem damage?

A. Carbon monoxide (CO)
B. Oxides of nitrogen (NO_x)
C. Carbon dioxide (CO_2)
D. Hydrocarbons

2. A vehicle engine undergoes incomplete combustion of fuel. Which combination of pollutants is most directly associated with this process according to the given content?

A. CO and hydrocarbons
B. CO_2 and SO_2
C. NO and CO_2
D. SO_2 and NO_2

3. Which factor mainly determines how long a plastic item takes to decompose?

A. Its color and size
B. The material and structure of the plastic
C. The temperature of the surrounding air only
D. The amount of water it contains

4. Which type of pharmaceutical waste includes medicines regulated by specific legislation because of their potential for abuse or addiction?

A. Hazardous medicines
B. Non-hazardous medicines
C. Controlled drugs
D. Waste containers

5. Which characteristic best defines Persistent Organic Pollutants (POPs)?

A. They rapidly decompose through biological processes.
B. They are resistant to chemical, biological, and photolytic degradation.
C. They are naturally produced only by plants.
D. They are easily removed from the environment by sunlight.

6. **Which of the following is an example of a persistent organic pollutant (POP) listed in the given content?**

A. DDT
B. Ethanol
C. Glucose
D. Acetic acid

7. **Which of the following is correctly matched with its common use?**

A. TCE (Trichloroethylene) — Industrial solvent and metal degreaser
B. Atrazine — Dry-cleaning solvent
C. PCE — Agricultural herbicide
D. PFAS — Fertilizer

8. **Why can polychlorinated hydrocarbons accumulate in fish, birds, and humans?**

A. They are highly soluble in water and are rapidly excreted.
B. They are insoluble in water but soluble in fats, allowing them to concentrate in fatty tissues.
C. They easily evaporate and are eliminated through respiration.
D. They rapidly degrade through biological processes.

9. Which statement best explains why the term "organic" does not necessarily mean that a chemical is natural or safe?

A. All organic chemicals are produced only by living organisms.
B. Synthetic organic chemicals are always more toxic than natural organic chemicals.
C. Both natural and synthetic substances can be organic, and safety depends on factors such as toxicity, concentration, exposure route, and duration.
D. Organic chemicals are safe as long as they contain carbon.

10. **Before synthetic pesticides became widely used, which of the following was used as a naturally derived insect poison?**

A. Pyrethrins obtained from pyrethrum flowers
B. Atrazine is produced synthetically
C. TCE used as an industrial solvent
D. PFAS used in water-resistant products

11. **Why can PCBs become highly concentrated in fish even though their concentration in water is negligible?**

A. PCBs are highly soluble in water and are rapidly absorbed by fish.
B. PCBs are readily soluble in fat and resistant to chemical and thermal degradation, allowing

them to persist and bioaccumulate in organisms.
C. PCBs rapidly decompose in water and are converted into nutrients.
D. PCBs evaporate from water and are directly produced by fish.

12. **Which set of properties is most useful for predicting whether an organic chemical will remain in water, attach to soil/sediment, or accumulate in organisms?**

A. Color, odor, and melting point
B. Water solubility, soil/sediment sorption coefficient, and bioconcentration
C. Boiling point, density, and viscosity
D. Acidity, alkalinity, and electrical conductivity

13. **Which of the following hydrocarbon fractions is mainly associated with the lighter components obtained during the fractional distillation of crude oil?**

A. C_1–C_5 gases
B. C_{20}–C_{70} heavy fractions
C. >C_{70} residue
D. C_{20}–C_{50} waxy fractions

14. **Why are synthetic surfactants generally preferred over soaps for cleaning in hard water?**

A. Synthetic surfactants are completely insoluble in water.
B. Synthetic surfactants do not form insoluble precipitates with Ca^{2+} and Mg^{2+}.
C. Synthetic surfactants contain only hydrophilic groups.
D. Synthetic surfactants cannot form micelles.

15. **Which statement correctly describes the cleansing action of soaps and synthetic surfactants?**

A. Their hydrophobic portions dissolve in water while their hydrophilic portions attach to grease.
B. Their hydrophilic and hydrophobic portions both dissolve equally in grease.
C. Their hydrophobic portions interact with oils, and their hydrophilic portions interact with water, allowing micelle formation.
D. They remove grease by chemically converting it directly into water

16. **A soap solution is used in mildly acidic water and shows reduced cleaning efficiency. Which explanation best accounts for this observation?**

A. Ca^{2+} and Mg^{2+} ions are completely removed from the solution.
B. The carboxylate head groups become protonated, reducing their polarity and ability to solubilize oils and grease.

C. The hydrocarbon chains become hydrophilic and dissolve completely in water.
D. The soap undergoes polymerization and loses its surfactant properties.

17. **Which combination correctly identifies the natural polymers and their building blocks?**

A. Proteins — sugars; cellulose — amino acids; DNA — fatty acids
B. Proteins — amino acids; starch and cellulose — sugars; DNA — nucleotides
C. Proteins — nucleotides; cellulose — amino acids; DNA — sugars
D. Proteins — hydrocarbons; starch — amino acids; DNA — fatty acids

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19. **Why does plastic waste persist in the environment for much longer than materials such as food, paper, and leather?**

A. Plastics contain strong covalent chemical bonds that are resistant to natural decomposition.
B. Plastics are highly soluble in water and therefore remain suspended for long periods.
C. Plastics are rapidly converted into simpler compounds by microorganisms.
D. Plastics contain naturally occurring nutrients that prevent decomposition.

20. **Which statement best explains why mismanaged plastic waste is considered a major environmental concern?**

A. Most plastic waste is immediately biodegraded, but the process releases excess oxygen.
B. Plastic production is increasing rapidly, while a large proportion of waste is landfilled or released into the environment, allowing persistent waste to accumulate and affect ecosystems.
C. Plastic waste decomposes faster than food and paper, causing short-term pollution only.
D. Nearly all plastic waste is recycled or converted into energy, so environmental accumulation is minimal.

21. **1. Which option represents the correct order of priority in the waste hierarchy, from the most preferred to the least preferred approach?**

A. Landfilling → Volume reduction → Recycling → Source reduction
B. Source reduction → Reuse & recycling → Composting → Volume reduction → Landfilling

C. Recycling → Landfilling → Composting → Source reduction
D. Composting → Landfilling → Reuse → Source reduction

22. What is the main purpose of the air-pollution control system in a waste-to-energy facility?

A. To increase the moisture content of waste
B. To remove pollutants from combustion gases before their release
C. To convert ash directly into electricity
D. To prevent steam formation in the boiler

23. A municipality wants to manage plastic waste according to the principles of waste minimization. Which strategy would best address the problem at its source?

A. Increase landfilling capacity for discarded plastics
B. Incinerate all plastic waste without further processing
C. Redesign products and reduce the use of unnecessary and single-use plastics
D. Increase the volume of plastic packaging to improve product protection

24. In a waste-to-energy facility, which sequence correctly describes the conversion of solid waste into electricity?

A. Waste combustion → water evaporation → steam → turbine → electricity
B. Waste combustion → electricity → steam → turbine → ash collection
C. Waste combustion → ash formation → water evaporation → turbine → electricity
D. Waste combustion → steam → water condensation → turbine → electricity

25. Why is landfilling considered the lowest-priority option in the waste-management hierarchy, even when modern landfills use liners and environmental controls?

A. Landfills completely eliminate the need for waste treatment.
B. Landfilling primarily isolates waste rather than preventing its generation, reuse, or recovery of materials and energy.
C. Landfills always release all waste directly into groundwater.
D. Landfilling converts all plastics into biodegradable materials.

26. The process requiring an enzyme system with iron and molybdenum-containing cofactors, by which the microorganism converts atmospheric nitrogen into a chemically usable nitrogen-containing compound, is called
(A). Denitrification, (B) Biological nitrogen fixation, (C). Nitrification, (D). Ammonification.

27. In the Haber-Bosch process ($N_2 + 3H_2 \rightleftharpoons 2NH_3$), The high pressure is used primarily because
(A). High pressure converts nitrogen into a more reactive form

(B). High pressure increases the activation energy of the reverse reaction
(C) The forward reaction produces fewer gaseous molecules than the reactants
(D). The forward reaction produces a greater number of gaseous molecules than the reactants.

28. **NO, in the nitrogen oxide pathway, is derived from**
(A) $NO_3^- \rightarrow NO_2^- \rightarrow NO$, (B). $N_2 \rightarrow NH_3 \rightarrow NO_3^- \rightarrow NO$, (C). $NH_3 \rightarrow NO_2^- \rightarrow N_2 \rightarrow NO$,
(D). $N_2 \rightarrow NO_2^- \rightarrow NO_3^- \rightarrow NO$.

29. **A low endothelial NO bioavailability would be expected to alter several processes involved in vascular homeostasis, except**
(A). Increased vascular smooth-muscle relaxation
(B). Increased proliferation of vascular smooth-muscle cells
(C). Decreased tendency toward atherosclerotic plaque development
(D). Decreased platelet aggregation

30. **Which of the following pollutant-category associations is correctly matched**
(A). Benzo[a]pyrene -Pesticides, (B). Aldrin- Polycyclic aromatic hydrocarbons (C) Chlorobiphenyl -Chemicals from industries, (D). Hexachlorodibenzofuran- Pesticides

31. **Two measurements were performed on the same absorbing species at the same wavelength and concentration, using the same incident radiation intensity. A 1 cm cell gives an absorbance of 0.18, and the second measurement is performed using a cell having three times the optical path length. While the system remains within the Beer-Lambert regime, the ratio of the transmitted intensities (P_2/P_1) will be:**
(A). 0.44, (B). 0.36, (C). 0.54, (D). 0.67

32. **Which pair contains one primary and one secondary atmospheric pollutant, respectively?**
(A). NO_2 and SO_2, (B). CO and SO_2, (C). NO_2 and CO, (D) CO and NO_2

33. **To obtain a stronger absorbance without increasing the pollutant concentration or changing the selected wavelength. Which change would achieve this?**
(A). Reduce the path length while keeping all other conditions unchanged
(B). Reduce the amount of radiation incident on the sample
(C). Dilute the sample while using the same path length
(D) Increase the path length while keeping all other conditions unchanged

34. **A spectrophotometric measurement gives an incident intensity $P_0 = 2.0$ mW and a transmitted intensity $P = 0.20$ mW. If the absorbance is defined as $A=\log(P_0/P)$, which value is closest to the absorbance?**
(A). 0.10, (B). 0.50, (C) 1.00, (D). 2.00

35. **Arrange the following in order of increasing CO stretching frequency:**
(A) $Ni(CO)_4 < Cr(CO)_6 < V(CO)_6 < Co_2(CO)_8$

(B) $[Ti(CO)_6]^{2-} < V(CO)_6 < Cr(CO)_6 < Ni(CO)_4$
(C) $Cr(CO)_6 < Ni(CO)_4 < Fe(CO)_5 < [Ti(CO)_6]^{2-}$
(D) $V(CO)_6 < Co_2(CO)_8 < Ni(CO)_4 < Cr(CO)_6$

36. **In haemoglobin and myoglobin, the Fe(II) of the heme is prevented from binding CO in a linear (most stable) fashion because of:**
(A) Oxidation of Fe(II) to Fe(III), (B)Steric clash with the distal histidine
(C) Absence of the proximal histidine, (D) Steric clash in the porphyrin ring being non-planar

37. **The fraction of total atmospheric CO consumption attributed to soil uptake is approximately, and a change in the balance between CO production and soil uptake can severely impact:**
(A) 40% and stratosphere, (B) 50% and stratosphere, (C) 40% and troposphere, (D) 40% and troposphere

38. **In a catalytic converter, the overall simplified reaction for simultaneous NO reduction and CO oxidation is:**
(A) $2NO + 2CO \rightarrow N_2 + 2CO_2$, (B) $NO + CO \rightarrow NO_2 + C$, (C) $2NO_2 + 2CO \rightarrow N_2O_4 + 2CO_2$, (D) $NO + O_2 \rightarrow NO_3$

39. **In the $BeCl_2$ molecule, which of the following is true:**
(A)Symmetric stretch is Raman active, IR inactive and Asymmetric stretch is Raman inactive, IR active
(B) Symmetric stretch is Raman inactive, IR active and Asymmetric stretch is Raman active, IR inactive
(C) Both symmetric and asymmetric stretches are IR active
(D) Neither stretching mode is IR or Raman active

40. **Nitrogen oxides (NO_x) are formed from atmospheric N_2 and O_2 primarily because:**
(A) N_2 and O_2 react spontaneously at normal atmospheric temperature (B) At high temperatures, N_2 and O_2 can react, (C) NO_x is formed only by biological nitrogen fixation
(D) O_2 reduces N_2 at the electrode surface in all engines

41. **Which of the following is a red-brown toxic gas and has an unpleasant acrid odour?**
(A) NO_3, (B) N_2O, (C) N_2O_5, (D)NO_2

42. **In oxymyoglobin, oxygen is coordinated between iron and imidazole side chain of:**
(A) His-62, (B) His-68, (C) His-64, (D) His-60

43. **Carbon monoxide poisoning of 6400 ppm in humans can cause:**
(A) Headache and dizziness, (B) Headache and loss of judgement,
(C) Unconsciousness and death, (D) Respiratory arrest and death

44. **Complete oxidation of octane: $2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$. The mass of CO_2 formed from 1.14 kg of octane is (C = 12, H = 1, O = 16)**

(A) 1.76 kg, (B) 4.40 kg, (C) 7.04 kg, (D) 3.52 kg

45. **In the atmosphere: $SO_2 + \bullet OH \rightarrow HSO_3\bullet$; $HSO_3\bullet + O_2 \rightarrow SO_3 + HOO\bullet$; $SO_3 + H_2O \rightarrow H_2SO_4$ (g); H_2SO_4 (g) + $H_2O \rightarrow H_2SO_4$ (aq). The net reaction is:**

(A) $2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$, with •OH regenerated

(B) $SO_2 + \bullet OH + O_2 + 2H_2O \rightarrow H_2SO_4$ (aq.) + HOO•

(C) $SO_2 + \bullet OH + H_2O \rightarrow H_2SO_4 + H\bullet$

(D) $SO_2 + 2\bullet OH \rightarrow H_2SO_4$

46. **Overall scheme: (1) $RCH_3 + 2O_2 + 2NO \rightarrow RCHO + 2NO_2 + H_2O$; (2) $2NO_2 + 2H_2O \rightarrow 2NO + 4\bullet OH$. For 100 molecules of RCH_3 going through one complete cycle, the number of •OH radicals formed and the net NO consumed are:**

(A) 400 and 0, (B) 400 and 200, (C) 200 and 200, (D) 200 and 0.

47. **Using bond dissociation energies (kJ/mol): CH_4 (427); C_2H_6 (406); C_3H_8 (primary C–H) (393); benzene (427); toluene (benzylic C–H) (326), the correct order of reactivity towards H-abstraction by •OH is:**

(A) Toluene > propane > ethane > benzene, (B) Benzene > methane > ethane > propane

(C) Methane > ethane > propane > toluene, (D) Toluene > benzene > propane > methane

48. **Acetaldehyde reacts with •OH, then O_2 , then NO_2 to form peroxyacetyl nitrate (PAN), $CH_3COOONO_2$. The maximum mass of PAN obtainable from 44 g of CH_3CHO is (C = 12, H = 1, N = 14, O = 16)**

(A) 121 g, (B) 105 g, (C) 137 g, (D) 76 g

49. **What is the amount of ozone over the Antarctic is?**

(A) 293.4 DU, (B) 220 DU, (C) 270 DU, (D) 250 DU

50. **For $O_2 + hv \rightarrow 2O$, $\Delta H = 498$ kJ/mol. The wavelength that can dissociate O_2 to a single oxygen atom is?**

(A) 120 nm, (B) 240 nm, (C) 480 nm, (D) 330 nm